

SECTION A(1) (35 marks)

1. Make m the subject of the formula $\frac{m}{2} - 1 = \frac{5m - n}{3}$. (3 marks)

2. Simplify $\frac{(uv^2)^5}{u^7v^{-3}}$ and express your answer with positive indices. (3 marks)

Answers written in the margins will not be marked.

3. The coordinates of the points P and Q are $(-6, 5)$ and $(2, 1)$ respectively. P is translated downwards by r units to R .

- (a) Write down, in terms of r , the coordinates of R .
 (b) Find r such that $\angle PQR = 90^\circ$.

(3 marks)

4. Factorize

- (a) $5x^2y + 17xy - 12y$,
 (b) $5x^2y + 17xy - 12y - 10x + 6$.

(4 marks)

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5. The sum of the number of blue pens and the number of red pens in a box is 795. If 123 blue pens are taken out from the box, then the number of blue pens in the box is 25% less than the number of red pens in the box. Find the number of red pens in the box. (4 marks)

6. Consider the compound inequality

$$3x - 4 < 5 \leq \frac{2 - 7x}{6} \dots\dots\dots (*)$$

- (a) Solve (*).
- (b) Write down the greatest integer satisfying (*).

(4 marks)

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7. A profit of \$162 is made by selling a ring at a discount of 15% on its marked price. The marked price of the ring is \$270 above its cost. Find the percentage profit. (4 marks)

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(a) By considering a pair of similar triangles, find the length of BC .

(b) Suppose that $AD = 10$ cm. Prove that $\triangle ABC \cong \triangle CEF$.

(5 marks)

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9. The frequency distribution table and the cumulative frequency distribution table below show the distribution of the weights of 20 apples.

Weight (grams)	Frequency
91 – 100	2
101 – 110	a
111 – 120	6
121 – 130	7
131 – 140	h

Weight less than (grams)	Cumulative frequency
100.5	2
110.5	6
120.5	12
130.5	b
140.5	k

- (a) Find a and b .
- (b) Find the mean of the distribution.

(5 marks)

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SECTION A(2) (35 marks)

10. The stem-and-leaf diagram below shows the distribution of the numbers of driving hours of a group of taxi drivers in a week.

Stem (tens)	Leaf (units)
2	5 6 9
3	2 3 3 6 6 7 8
4	0 0 2 2 5 5 7 7 7
5	w

The ratio of the range to the inter-quartile range of the distribution is $7:3$.

- (a) Find w . (3 marks)
- (b) (i) Write down the mode of the distribution.
- (ii) If two taxi drivers are randomly selected from the group, find the probability that the mean of their numbers of driving hours in the week exceeds the mode of the distribution. (3 marks)

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12. The straight line L passes through the point $(-10, 11)$. The slope of L is -2 .

(a) Find the equation of L . (2 marks)

(b) The coordinates of the points H and K are $(2, 5)$ and $(-4, -7)$ respectively. Let P be a point lying on L such that P is equidistant from H and K .

(i) Find the coordinates of P .

(ii) Let Q be a point such that $HPKQ$ is a quadrilateral, where $HP = HQ = KQ$. Find the area of the quadrilateral $HPKQ$. (5 marks)

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13. A solid metal sphere of radius 10 cm is melted and recast into three similar solid right circular cylinders X , Y and Z . Let r_1 cm, r_2 cm and r_3 cm be the base radii of X , Y and Z respectively. It is given that $r_1 : r_2 : r_3 = 2 : 3 : 5$.

(a) Express the volume of Y in terms of π . (3 marks)

(b) A craftsman finds that the height of Y is 9 cm.

(i) Find r_2 .

(ii) The craftsman claims that the total surface area of Z is greater than the surface area of the sphere. Do you agree? Explain your answer.

(5 marks)

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15. A bag contains 3 red balls, 3 white balls and 5 black balls. If 4 balls are randomly drawn from the bag at the same time, find

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16. It is given that P , Q and R are three points in a rectangular coordinate plane. The coordinates of P and Q are $(7, 8)$ and $(-13, 20)$ respectively. The equation of the perpendicular bisector of PR is $3x + 5y - 27 = 0$. Let C be the circle which passes through P , Q and R .

- (a) Find the equation of C . (3 marks)
- (b) The tangent to C at P and the tangent to C at R intersect at the point H . Is $\triangle HPR$ a right-angled triangle? Explain your answer. (3 marks)

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(a) Find

(i) the distance between A and C ,

(ii) $\angle BAD$.

(4 marks)

(b) It is given that $VD = 80$ cm . A student finds that the angle between the plane $ABCD$ and the horizontal ground is 65° . The student claims that the angle between VD and the horizontal ground is less than 50° . Is the claim correct? Explain your answer. (2 marks)

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18. Let $G(n)$ be the n th term of a geometric sequence. It is given that $G(2) = 12$ and $G(5) = 768$.

(a) Find $G(1)$. (2 marks)

(b) Define $T(n) = \log_2 G(n)$ for all positive integers n . Find

(i) $T(n+1) - T(n)$,

(ii) the greatest value of k such that $T(1) + T(2) + T(3) + \dots + T(k) < 10^6$.

(5 marks)

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19. Let $h(x) = -x^2 + 30kx + 4x - 225k^2 - 40k + 3$, where k is a positive constant. On the same rectangular coordinate system, denote the vertex of the graph of $y = h(x)$ and the vertex of the graph of $y = 14 - h(4 - x)$ by A and B respectively.

(a) Using the method of completing the square, express, in terms of k , the coordinates of A . (2 marks)

(b) Write down, in terms of k , the coordinates of B . (1 mark)

(c) The coordinates of the point C are $(2 + 48k, 7 - 36k)$. Let I and J be the in-centre and the circumcentre of $\triangle ABC$ respectively.

(i) Let M be the mid-point of AB . Describe the geometric relationship between CM and AB . Explain your answer.

(ii) Express, in terms of k , the distance between I and J .

(iii) The coordinates of the point D are $(6 - 22k, 6 - 15k)$. Is it possible that $BDIJ$ is a parallelogram? Explain your answer. (9 marks)

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